Networking Task 02: Network Devices & IP Addressing

Part A: Network Devices Research

Router

Purpose:
A router is a networking device that connects different networks together, such as a home network and the internet. It allows multiple devices to access the internet through a single connection

How it works:
The router receives data from devices connected to the network and forwards it to the correct destination using IP addresses. It also receives data from the internet and sends it to the correct device within the network

Real-world usage:
Routers are commonly used in homes, offices, schools, and colleges. The Wi-Fi router at home is a common example that connects phones, laptops, smart TVs, and other devices to the internet

Switch

Purpose:
A switch is used to connect multiple devices within the same network. It helps devices communicate with each other efficiently

How it works:
A switch uses MAC addresses to identify devices connected to it. When data is received, it sends the data only to the intended device instead of broadcasting it to everyone

Real-world usage:
Switches are widely used in offices, computer labs, and organizations where many computers need to be connected to the same network

Hub

Purpose:
A hub is a basic networking device used to connect multiple devices in a network

How it works:
Unlike a switch, a hub does not identify the destination device. It simply sends incoming data to all connected devices, whether they need it or not

Real-world usage:
Hubs were commonly used in older networks but have mostly been replaced by switches because switches are faster, more efficient, and more secure

Access Point

Purpose:
An Access Point (AP) provides wireless connectivity to devices and extends the coverage of a network

How it works:
It connects to a wired network and creates a Wi-Fi signal that allows wireless devices such as laptops and smartphones to access the network

Real-world usage:
Access points are commonly found in colleges, airports, shopping malls, hotels, and offices where many users need wireless internet access

Firewall

Purpose:
A firewall is a security device or software that protects a network from unauthorized access and cyber threats

How it works:
It monitors incoming and outgoing network traffic and applies security rules to determine whether data should be allowed or blocked. This helps prevent malicious traffic from entering the network

Real-world usage:
Firewalls are used in homes, businesses, government organizations, and data centers to protect systems from hackers, malware, and other cyber attacks

Modem

Purpose:
A modem is a device that connects a computer network to an Internet Service Provider (ISP), making internet access possible

How it works:
It converts signals received from the ISP into digital signals that computers and other devices can understand. It also converts outgoing digital signals for transmission over the internet connection

Real-world usage:
Modems are used in homes and offices with broadband, cable, DSL, or fiber internet connections. They are often connected to a router to provide internet access to multiple devices

Part B: IP Address Classification

1. 192.168.1.10 - Private IP

This IP address belongs to the 192.168.0.0 – 192.168.255.255 private IP range. Private IP addresses are used within local networks such as homes, schools, and offices and cannot be accessed directly from the internet.

2. 10.0.0.5 - Private IP

This address falls within the 10.0.0.0 – 10.255.255.255 private IP range. It is commonly used in organizations and private networks for internal communication between devices.

3. 172.16.5.20 -Private IP

This IP belongs to the 172.16.0.0 – 172.31.255.255 private IP range. Since it falls within this range, it is considered a private IP and is used for internal network communication.

4. 8.8.8.8 - Public IP

This is a public IP address. It is owned by Google and is used as a public DNS server. Public IP addresses are accessible over the internet and are globally unique.

5. 1.1.1.1 - Public IP

This is a public IP address operated by Cloudflare. It is a publicly accessible DNS server and can be reached from anywhere on the internet.

6. 192.168.100.1 - Private IP

This IP address falls within the 192.168.x.x private IP range. It is commonly used as the default gateway or management address of routers and modems in local networks

Part C: Understanding Your Network

1. Which IP range does your device belong to?

My device belongs to the 192.168.1.x range, which is part of the 192.168.0.0 – 192.168.255.255 private IP address range. This range is commonly used in home and office networks.

2. Is it Public or Private?

The IP address of my device is a Private IP Address. Private IP addresses are used within local networks and cannot be accessed directly from the internet.

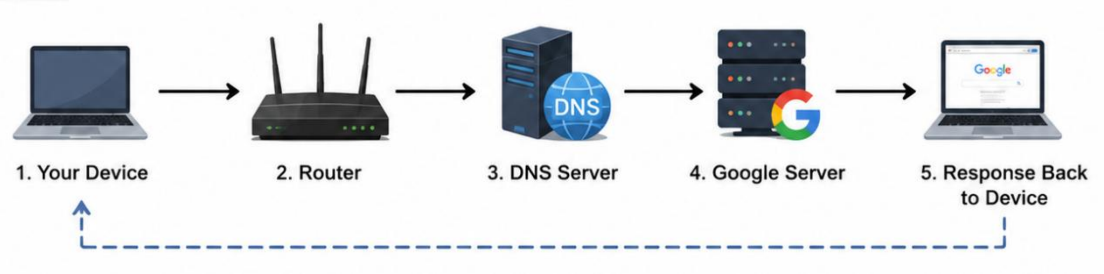
3. What role does your router play in your network?

The router acts as a bridge between my local network and the internet. It manages network traffic, assigns IP addresses to devices, and ensures that data reaches the correct destination.

4. What would happen if the DNS server stopped working?

If the DNS server stopped working, I would not be able to access websites using their domain names, such as google.com. Internet access might still work through IP addresses, but website names would not be translated into IP addresses.

Part D: Network Communication Flow



1. Your Device

When I enter www.google.com in my web browser and press Enter, my device starts a request to access the website. Since computers communicate using IP addresses, my device first needs to find the IP address associated with the domain name.

2. Router

The request is sent to the router which acts as the gateway between my local network and the internet. The router forwards the request to the appropriate DNS server so the website's IP address can be identified.

3. DNS Server

The DNS (Domain Name System) server receives the request and translates the domain name www.google.com into its corresponding IP address. It then sends this IP address back so the device knows where to send the request.

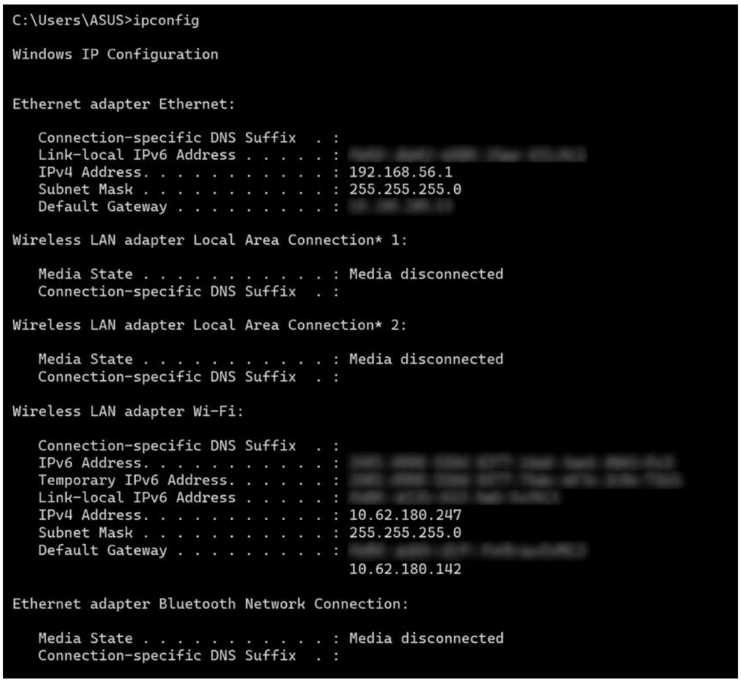
4. Google Server

Using the IP address provided by the DNS server, the request reaches Google's server. The server processes the request and gathers the required webpage data, such as text, images, and other content.

5. Response Back to Device

Google's server sends the requested webpage data back through the internet to the router and then to my device. The web browser receives the data and displays the Google homepage on the screen.

Part E: Practical Command Exercise



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C:\Users\ASUS>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : [illegible]
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : [illegible]

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : [illegible]
   Temporary IPv6 Address. . . . . . : [illegible]
   Link-local IPv6 Address . . . . . : [illegible]
   IPv4 Address. . . . . . . . . . . : 10.62.180.247
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : [illegible]
                                       10.62.180.142

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
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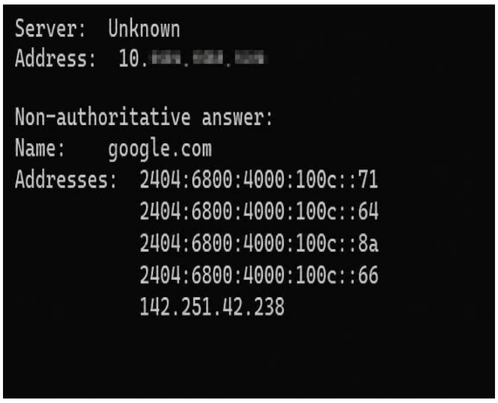

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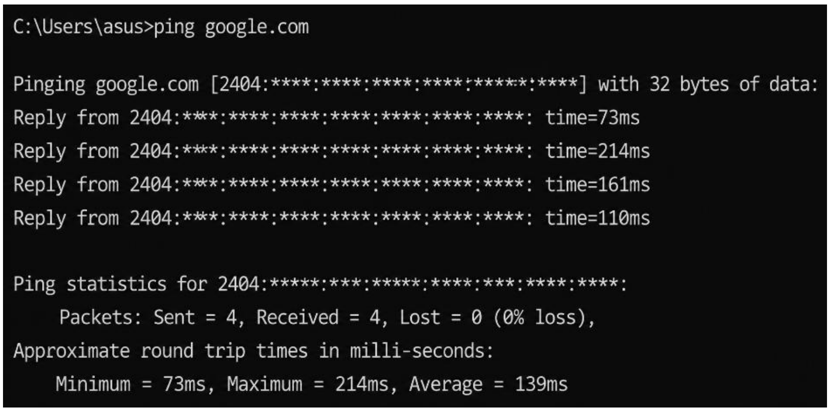
Server:  Unknown
Address:  10.[illegible]

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4000:100c::71
          2404:6800:4000:100c::64
          2404:6800:4000:100c::8a
          2404:6800:4000:100c::66
          142.251.42.238
```



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C:\Users\asus>ping google.com

Pinging google.com [2404:****:****:****:****:*****:****] with 32 bytes of data:
Reply from 2404:****:****:****:****:****:****:****: time=73ms
Reply from 2404:****:****:****:****:****:****:****: time=214ms
Reply from 2404:****:****:****:****:****:****:****: time=161ms
Reply from 2404:****:****:****:****:****:****:****: time=110ms

Ping statistics for 2404:*****:***:*****:****:***:****:****:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 73ms, Maximum = 214ms, Average = 139ms
```

1. What IP address did DNS return for Google?

The DNS server returned the IP address 142.251.xx.xxx for google.com.

2. Was the ping successful?

Yes, the ping was successful. The device was able to communicate with Google's server and received responses successfully.

3. Why is DNS important before communication begins?

DNS is important because it converts domain names such as google.com into IP addresses that computers can understand. Without DNS, users would have to remember and enter numerical IP addresses instead of website names.